National Institute of Technology Kurukshetra
Department of Computer Engineering

Problems Solving and Programming using C (CSIC101)

Mid-I Exam, Sept-2024

Timing: 50 Min. | Sept-2024. Semester-I | Max Marks: 20

NOTE: There is no syntax and syntactic errors in the source code provided in the question paper.

1. (a) A five-digit number is entered through the keyboard. Write a C program to obtain the reversed number and to determine whether the original and reversed numbers are equal or not. **(5)**

   (b) Write a C program to receive value of an angle in degrees and check whether sum of squares of sine and cosine of this angle is equal to 1.You can use only conditional operators with if-else. **(5)**

2. (a) State the order of evaluation of the operators in each of the following C statements and show the value of x after each statement is performed.
   $x = (3 * 9(*(3 + (9 * 3/(3))))$ **(2)**

   (b) Wind chill factor is the felt air temperature on exposed skin due to wind. The wind chill temperature is always lower than the air temperature, and is calculated as per the following formula:
   $wcf = 35.74 + 0.6215t + (0.4275t - 35.75) * \exp v^{0.16}$
   where **t** is the temperature and **v** is the wind velocity. Write a C program to receive values of **t** and **v** and calculate wind chill factor **(wcf)**. **(4)**

3. Write the output for the given C programs. **(4)**

```c
#include <stdio.h>
int main( )
{
    int i = -1, j = 1, k, l;
    k = !i && j;
    l = !i || j;
    printf ( "%d %d\n", k, l);
    return 0;
}
```

```c
#include<stdio.h>
int main()
{
    int j = 65 ;
    printf ( "j >= 65 ? %d : %c\n", j ) ;
    return 0 ;
}
```